Ex No:4	Visualization and Analysis of Student Social Media Usage Data
Date	

Aim:

To analyze student social media usage and addiction data using appropriate data visualization techniques in order to identify behavioral patterns, academic impact, and health-related trends.

Software Used

• Programming Language: Python 3.x

• Libraries:

- o Pandas (for data loading and manipulation)
- o Matplotlib (for creating visualizations)
- o Seaborn (for statistical and multivariate visualizations)

• Environment: Jupyter Notebook.

Description:

Social media usage among students has increased significantly, influencing academic performance, sleep patterns, and mental well-being. Visual analysis of such data helps in understanding behavioral trends and identifying correlations between usage habits and health or academic indicators.

In this experiment, a real-world dataset containing student social media addiction attributes is analyzed using multiple visualization techniques. The dataset includes demographic information, usage patterns, addiction scores, sleep duration, mental health scores, and academic impact indicators.

The following visualization methods are applied:

• **Bar Charts** to compare categorical frequencies such as platform usage and gender distribution.

• **Pie Charts** to represent proportions of academic levels and academic impact.

• **Histograms** to study the distribution of sleep hours.

• **Line Charts** to observe trends between usage duration and sleep hours.

• **Scatter Plots** to analyze relationships between sleep and mental health.

• **Box Plots** to identify variation and outliers in addiction scores.

• **Pair Plots** to explore multivariate relationships among numeric attributes.

• **Heatmaps** to visualize correlation strength between numerical features.

These visualizations collectively provide insights into student social media behavior, addiction patterns, and their possible effects on academics and health.

Dataset details

The dataset consists of student-level social media usage and behavioral attributes.

Demographic Attributes

• **Student_ID –** Unique identifier for each student

• **Age –** Age of the student in years

• **Gender** – Male or Female

• **Academic_Level** – High School, Undergraduate, or Graduate

• **Country** – Country of residence

Social Media Usage Attributes

• **Avg_Daily_Usage_Hours** – Average number of hours spent on social media per day

• **Most_Used_Platform** – Platform used most frequently

• **Addicted_Score** – Addiction level measured on a scale of 1–10

Academic and Health Indicators

• **Affects_Academic_Performance** – Whether social media negatively affects academics (Yes/No)

• **Sleep_Hours_Per_Night** – Average hours of sleep per night

• **Mental_Health_Score** – Self-reported mental well-being score (1–10)

• **Conflicts_Over_Social_Media** – Number of conflicts caused by social media usage

• **Relationship_Status** – Single, In Relationship, or Complicated

List of Tasks

1. Load the student social media addiction dataset into a Pandas DataFrame.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df=pd.read_csv("Students Social Media Addiction.csv")
df.head()
```

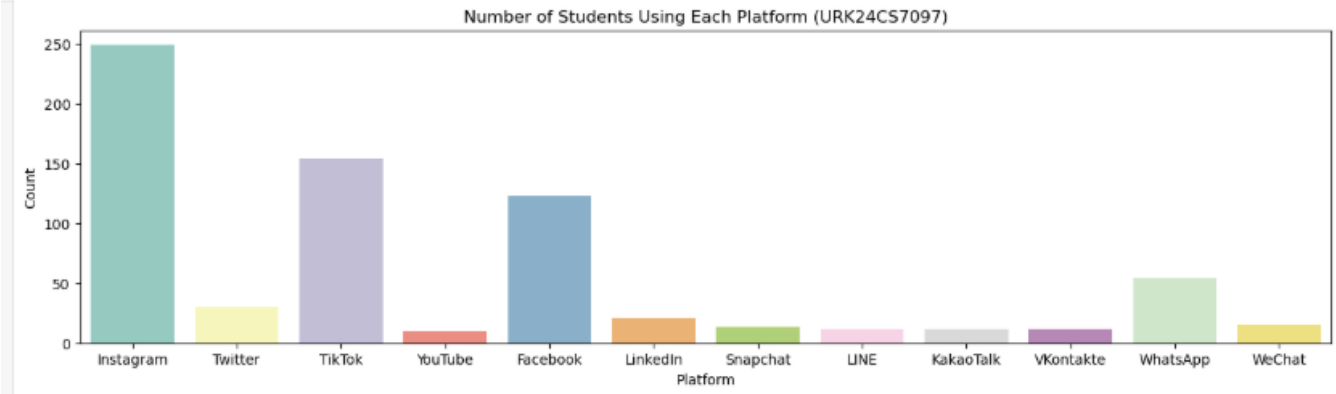
OUTPUT:

```
   Student_ID  Age  Gender Academic_Level     Country  Avg_Daily_Usage_Hours  \
0           1   19  Female  Undergraduate  Bangladesh                    5.2
1           2   22    Male       Graduate       India                    2.1
2           3   20  Female  Undergraduate         USA                    6.0
3           4   18    Male    High School          UK                    3.0
4           5   21    Male       Graduate      Canada                    4.5

  Most_Used_Platform Affects_Academic_Performance  Sleep_Hours_Per_Night  \
0          Instagram                          Yes                    6.5
1            Twitter                           No                    7.5
2             TikTok                          Yes                    5.0
3            YouTube                           No                    7.0
4           Facebook                          Yes                    6.0

   Mental_Health_Score Relationship_Status  Conflicts_Over_Social_Media  \
0                    6     In Relationship                            3
1                    8              Single                            0
2                    5         Complicated                            4
3                    7              Single                            1
4                    6     In Relationship                            2

   Addicted_Score
0               8
1               3
2               9
3               4
4               7
```

2. Create a bar plot to visualize the number of students using each social media platform.

```
plt.figure(figsize=(8,5))
sns.countplot(data=df, x="Most_Used_Platform")
plt.title("Number of Students by Most Used Social Media Platform")
plt.xlabel("Social Media Platform")
plt.ylabel("Number of Students")
plt.xticks(rotation=45,ha='right')
plt.show()
```

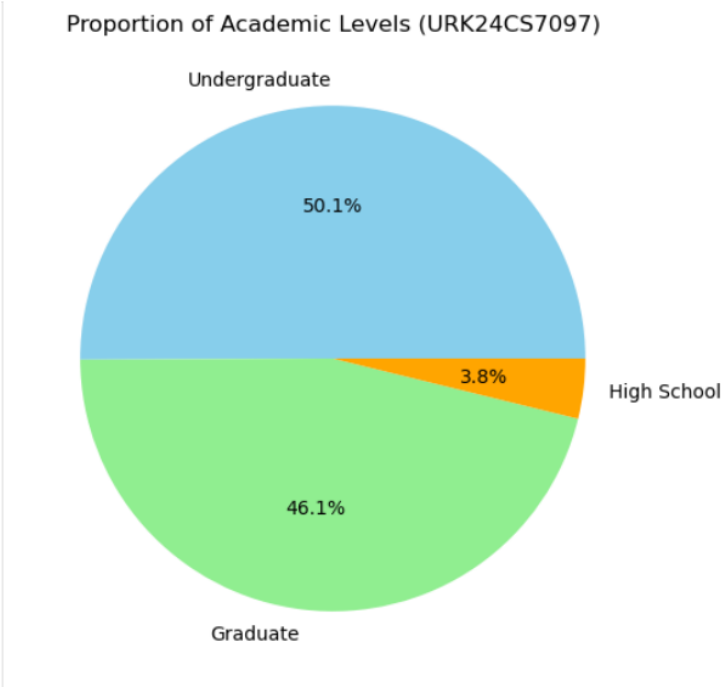
OUTPUT:



3. Plot a pie chart showing the proportion of students in each academic level.

```
academic_counts = df["Academic_Level"].value_counts()
plt.figure(figsize=(4,4))
plt.pie(academic_counts, labels=academic_counts.index, autopct="%1.1f%%", startangle=140)
plt.title("Academic Level Distribution")
plt.show()
```

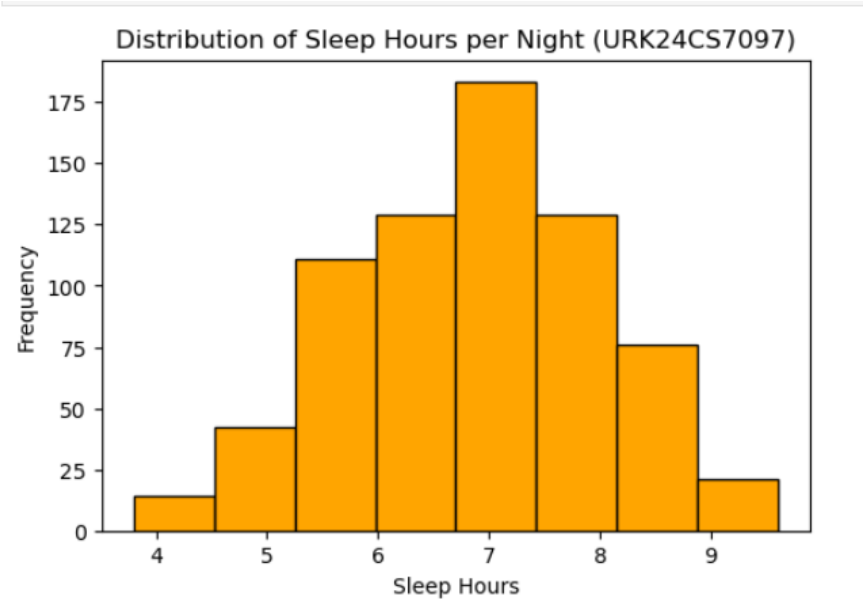
OUTPUT:



4. Draw a histogram to analyze the distribution of sleep hours per night.

```
plt.figure(figsize=(8,5))
plt.hist(df["Sleep_Hours_Per_Night"],bins=5, color="skyblue", edgecolor="black")
plt.title("Distribution of Sleep Hours per Night")
plt.xlabel("Sleep Hours")
plt.ylabel("Frequency")
plt.show()
```

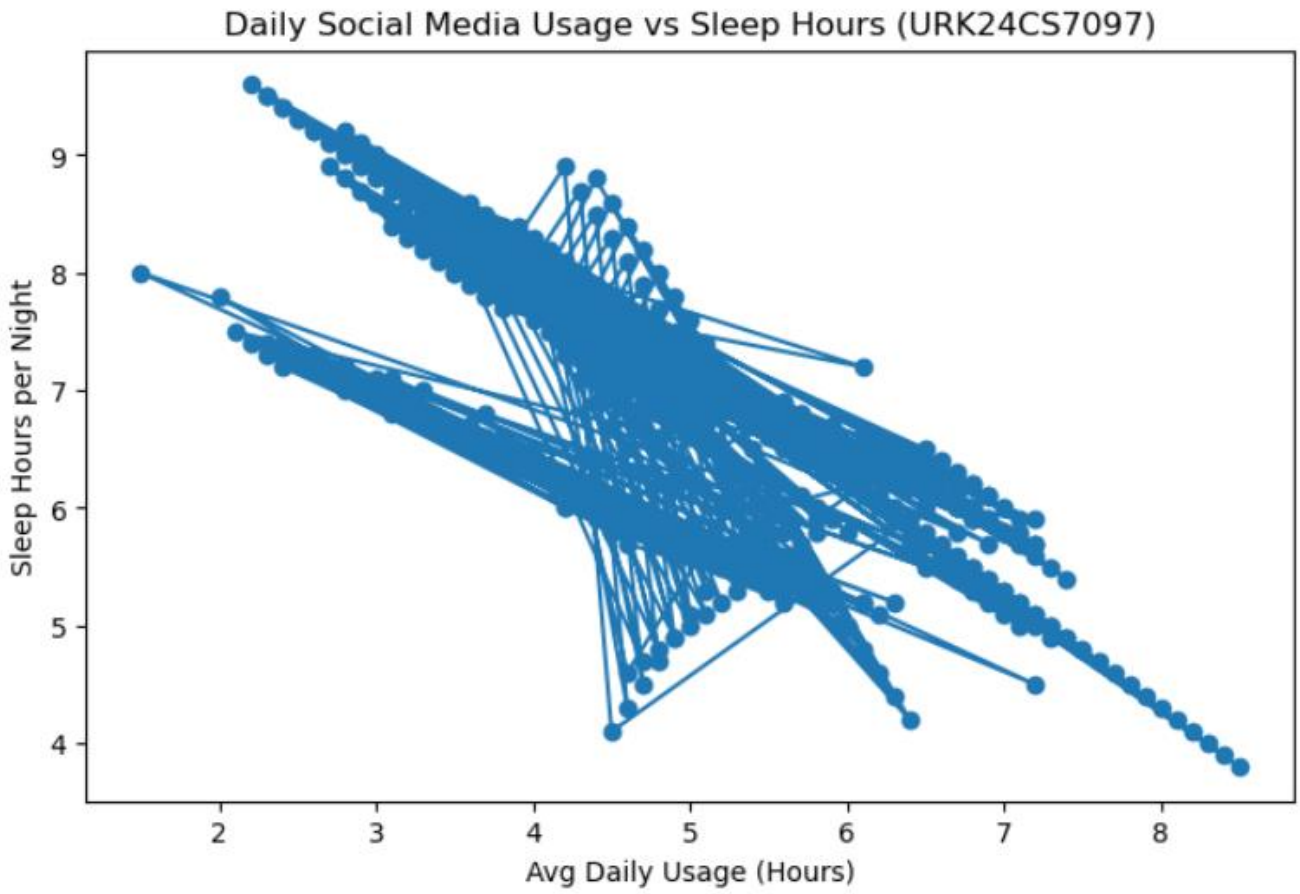
OUTPUT:



5. Use a line chart to examine the relationship between average daily social media usage and sleep hours.

```
usage_sleep = df.sort_values("Avg_Daily_Usage_Hours")
plt.figure(figsize=(8,5))
plt.plot(usage_sleep["Avg_Daily_Usage_Hours"], usage_sleep["Sleep_Hours_Per_Night"])
plt.title("Average Daily Social Media Usage vs Sleep Hours")
plt.xlabel("Average Daily Usage (Hours)")
plt.ylabel("Sleep Hours per Night")
plt.show()
```

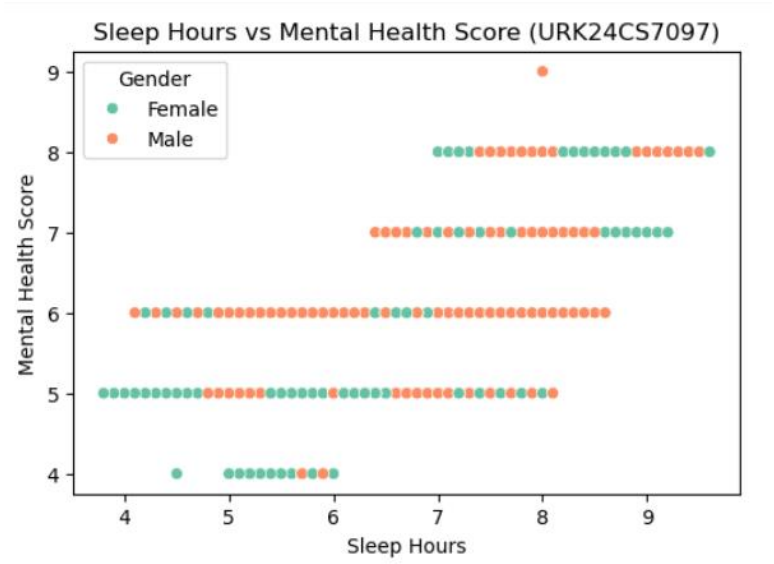
OUTPUT:



6. Plot a scatter diagram to study the relationship between sleep hours and mental health score, and interpret the correlation.

```
plt.figure(figsize=(8,5))
sns.scatterplot(data=df, x="Sleep_Hours_Per_Night", y="Mental_Health_Score")
plt.title("Sleep Hours vs Mental Health Score")
plt.xlabel("Sleep Hours per Night")
plt.ylabel("Mental Health Score")
plt.show()
```

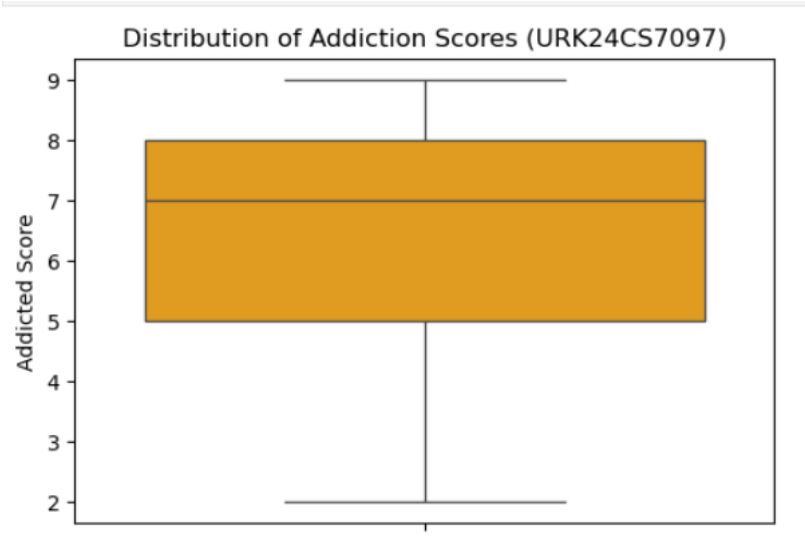
OUTPUT:



7. Create a box plot to analyze the distribution, median, and outliers of addiction scores.

```
plt.figure(figsize=(6,5))
sns.boxplot(y=df["Addicted_Score"])
plt.title("Distribution of Social Media Addiction Scores")
plt.ylabel("Addiction Score")
plt.show()
```

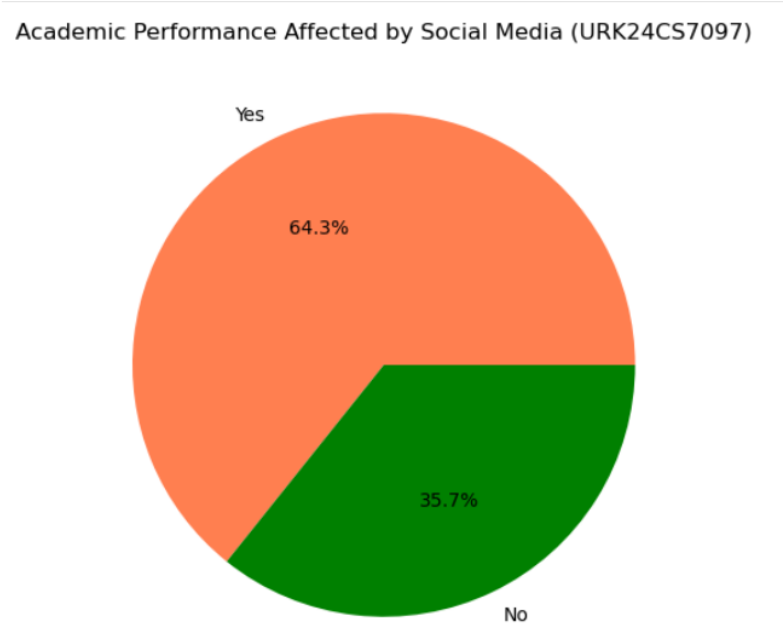
OUTPUT:



8. Plot a pie chart to represent the proportion of students whose academic performance is affected by social media usage.

```
impact_counts = df["Affects_Academic_Performance"].value_counts()
plt.figure(figsize=(4,4))
plt.pie(impact_counts, labels=impact_counts.index, autopct="%1.1f%%", startangle=140)
plt.title("Impact of Social Media on Academic Performance")
plt.show()
```

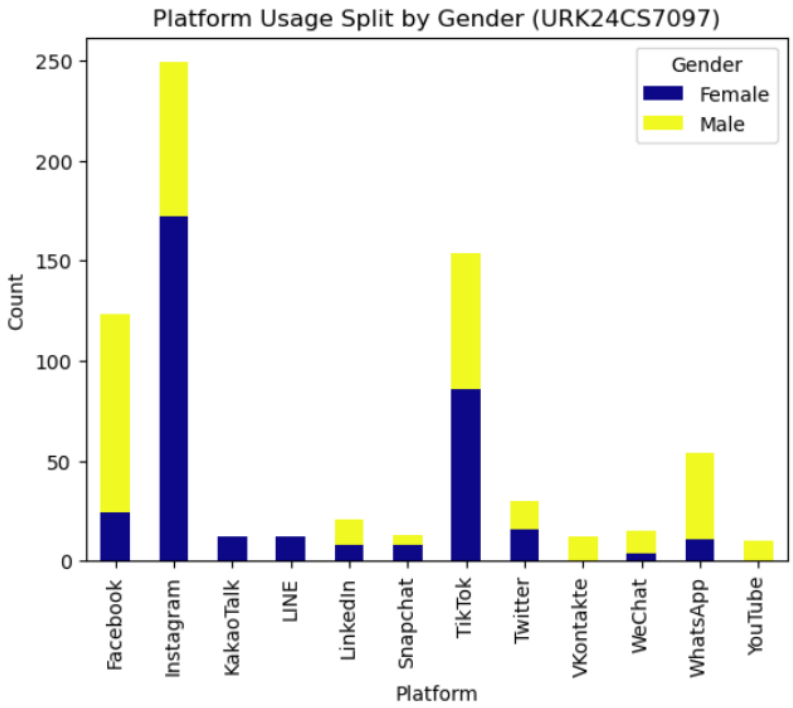
OUTPUT:



9. Generate a stacked bar chart showing social media platform usage split by gender.

```
platform_gender = pd.crosstab(df["Most_Used_Platform"], df["Gender"])
platform_gender.plot(kind="bar", stacked=True, figsize=(8,5))
plt.title("Social Media Platform Usage by Gender")
plt.xlabel("Platform")
plt.ylabel("Number of Students")
plt.show()
```

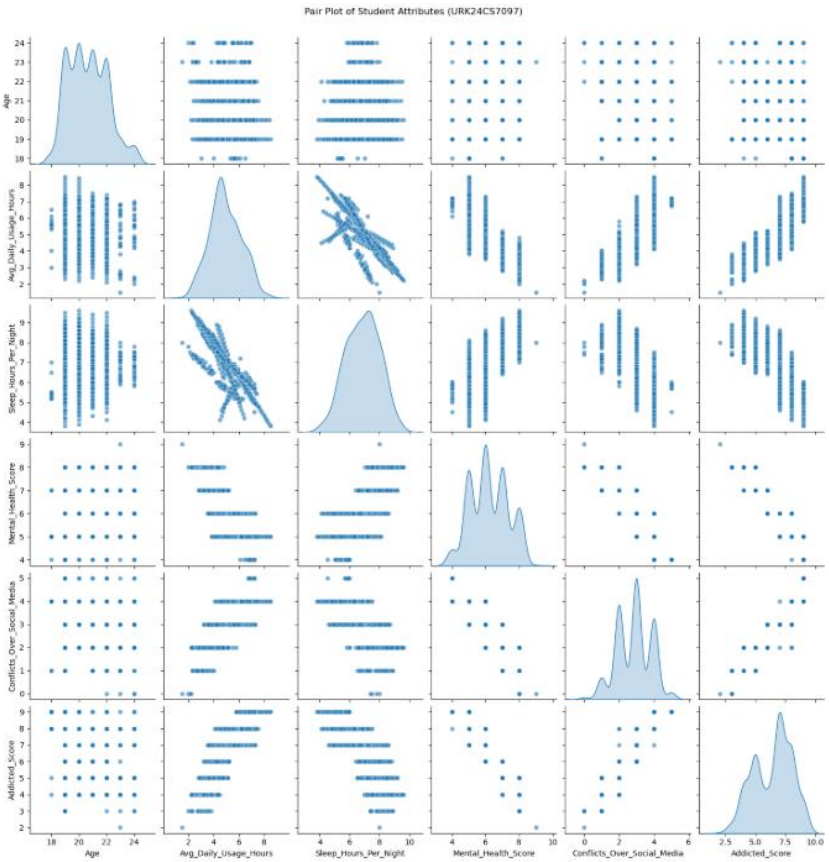
OUTPUT:



10. Use a pair plot to explore relationships between age, usage hours, sleep duration, mental health score, conflicts, and addiction score.

```
sns.pairplot(
    df[[
        "Age",
        "Avg_Daily_Usage_Hours",
        "Sleep_Hours_Per_Night",
        "Mental_Health_Score",
        "Conflicts_Over_Social_Media",
        "Addicted_Score"
    ]]
)
plt.show()
```

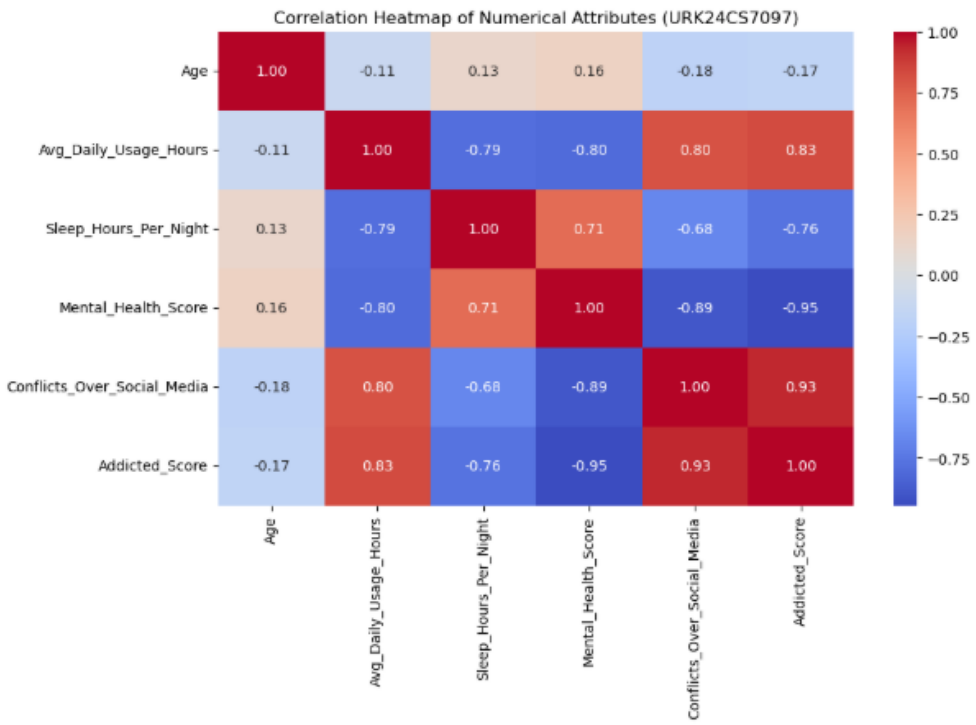
OUTPUT:



11. Create a correlation heatmap to visualize the relationships between all numerical attributes.

```
plt.figure(figsize=(5,6))
correlation_matrix = df.select_dtypes(include="number").corr()
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap of Numerical Attributes")
plt.show()
```

OUTPUT:



12. Interpret the visualizations and summarize the key insights obtained from the analysis.

Video Link –

https://youtu.be/JVKJCukRAyk

RESULT: Based on Visualization and Analysis of Student Social Media Usage Data the programs have been executed and outputs are executed successfully.